MATH 201 Your name: ______________________

Learning target IN3 and IN5, version 1

Evaluate a definite integral using the Fundamental Theorem of Calculus.

Explain how to compute the exact value of each of the following definite integrals using the Fundamental Theorem of Calculus.
Leave all answers in exact form, with no decimal approximations.

(a) $\displaystyle\int_{x=4}^{6} \left(\frac{6}{x}\right) dx$

(b) $\int_{x=\frac{5}{4}\pi}^{\frac{4}{3}\pi} \left(-2\sec^2(x)\right) dx$

(c) $\int_{x=-2}^{-1} \left(3x^3 + 10x - 1\right) dx$

Learning target IN3 and IN5, version 2

Evaluate a definite integral using the Fundamental Theorem of Calculus.

Explain how to compute the exact value of each of the following definite integrals using the Fundamental Theorem of Calculus.
Leave all answers in exact form, with no decimal approximations.

(a) $\displaystyle\int_{x=1}^{5} \left(\frac{2}{x} + 2^x\right) dx$

(b) $\displaystyle\int_{x=\pi/3}^{3\pi/4} (7\cos(x))\, dx$

(c) $\displaystyle\int_{x=-3}^{1} \left(9x^3 - 9x^2 + 2\right) dx$

MATH 201 Your name: ____________________

Learning target IN3 and IN5, version 3

Evaluate a definite integral using the Fundamental Theorem of Calculus.

Explain how to compute the exact value of each of the following definite integrals using the Fundamental Theorem of Calculus.

Leave all answers in exact form, with no decimal approximations.

(a) $\displaystyle\int_{x=-2}^{4} \left(-3x^2+4x-12\right) dx$

(b) $\displaystyle\int_{x=2}^{3} \left(e^x - \frac{3}{x}\right) dx$

(c) $\displaystyle\int_{x=\pi/6}^{2\pi/3} (4\sin(x))\, dx$

Learning target IN3 and IN5, version 4

Evaluate a definite integral using the Fundamental Theorem of Calculus.

Explain how to compute the exact value of each of the following definite integrals using the Fundamental Theorem of Calculus.
Leave all answers in exact form, with no decimal approximations.

(a) $\displaystyle\int_{x=0}^{\pi} \left(7\sin(x) - 2x^2\right) dx$

(b) $\displaystyle\int_{x=2}^{10} \left(\frac{3}{x} + 7^x\right) dx$

(c) $\displaystyle\int_{x=-1}^{3} \left(9x^4 - 12x^2 + 7x - 3\right) dx$